

Requirements Specification

Coffee Break Games – Memory Game

SEDA Masters Project

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1 Document Control and Purpose

This document provides the customer requirements specification for a networked two-player memory card game developed as part of the SEDA masters software engineering project. It follows the lightweight Volere-style structure with explicit fit criteria provided by the project tutor.

1.1 Project Goal

Deliver a two-player memory card game as a standalone executable Java application in which players take turns flipping cards on a shared board, attempting to find matching K-tuples of cards. The player who collects the most points by the end of the game wins. The game supports both local play on a single device and networked play between two devices on the same local area network.

1.2 Stakeholders

- Office employees playing against each other during breaks.
- Course assessors evaluating functional correctness and code quality.

1.3 Clarifications and Resolutions

The following points were clarified during requirements gathering and are recorded here for traceability.

- **Scoreboard:** The customer requested score visibility during and after the game. *Resolution:* a live in-game score display showing both players' current scores is required at all times during gameplay. A persistent cross-session leaderboard is not required.
- **K-tuple input and deck size options:** K is entered first by the player. The system then derives up to three valid deck size options dynamically as the largest multiples of K that do not exceed 45. The player selects from these computed options. See FR-MENU-04 and FR-MENU-05.
- **15-card deck and 15-symbol minimum:** Depending on the chosen K, deck sizes smaller than 30 may produce fewer than 15 distinct symbols. The 15-symbol minimum is a lower bound applied where mathematically achievable given the chosen K and deck size combination. See FR-VAL-03.
- **Turn continuation on match:** When a player successfully completes a K-tuple match, the turn remains with that player. The turn passes to the opponent only on a mismatch. See FR-FLIP-02 and FR-FLIP-03.
- **Real-time network play:** "Real time" in the context of LAN play is defined as a card flip being visible on the remote client within 500 ms of the active player's click. See QR-PERF-02.

2 Scope and Context

2.1 In Scope

- Main menu with game mode selection, configuration options, and player name entry.
- Two-player local gameplay on a single device.
- Two-player LAN gameplay between two devices on the same local area network, including host and join flows.
- LAN lobby screen showing connection status and both player names before the game starts.
- Card flip mechanics, K-tuple matching, and live score display.
- Dynamic deck size options derived from the player's chosen K value.
- Real-time board synchronisation between both clients during LAN play.
- Final result screen showing scores and win / lose / draw outcome.
- Option to restart or start a new game after a game session ends.
- Handling of mid-game network disconnections.

2.2 Out of Scope

- Single-player or AI opponent mode.
- Online play over the internet or across different networks.
- Automatic network discovery; the joining player must enter the host IP address manually.
- Card shuffling (card order is fixed or pre-determined).
- Persistent cross-session player accounts or historical leaderboard.
- A countdown timer during gameplay (may be added as an extension if time permits).
- Accumulated points carry-over between separate game sessions (may be added as an extension if time permits).
- Mobile or web-based clients.

2.3 Assumptions and Dependencies

- For local play, both players share one machine and take turns at the same input device.
- For LAN play, both machines are connected to the same local area network and can reach each other by IP address.
- Each machine is capable of running Java 24 or higher.
- The game is operated entirely via mouse; no keyboard input is required during gameplay (text entry fields for names, K value, and IP address may use the keyboard).
- Card images are bundled inside the `.jar` file; no external assets are required at runtime.
- There is no restriction on the visual style of card symbols; any distinct images are acceptable.
- External hyperlinks from within the application are permitted.

2.4 Glossary

- **Card:** A tile on the game board displaying either a face-down back or a face-up symbol image.
- **Symbol:** A unique image that appears on exactly K cards in the deck.
- **K-tuple:** A set of K cards sharing the same symbol; collecting one awards 10 points to the active player.
- **K:** The configurable number of matching cards required to form a valid tuple. K is a positive integer in the range 1–45. K = 1 is a valid (trivial) mode in which every flipped card is immediately a match.
- **Deck:** The complete set of cards laid out on the board for one game session.
- **Turn:** The period during which one player may flip cards. The turn continues as long as the player keeps forming matches; it ends only when a mismatch occurs.
- **Active player:** The player whose turn it currently is.
- **Mismatch:** The result when K flipped cards do not all share the same symbol. Causes the turn to pass to the opponent.
- **Match:** The result when all K flipped cards share the same symbol. Awards 10 points and keeps the turn with the active player.
- **Mismatch delay:** A configurable pause after a mismatch during which both players can see the revealed cards before they flip back face-down.
- **MISMATCH_DELAY_MS:** A named constant in the source code controlling the mismatch delay duration; default value 1500 ms.
- **Host:** The player who creates and serves a LAN game session.
- **Client:** The player who connects to an existing LAN game session hosted by another machine.
- **Lobby screen:** A pre-game screen displayed on both machines after a successful LAN connection, confirming both player names and connection status before the game board is shown.

3 Constraints

- **CON-TECH-01 (Technology):** The implementation language shall be Java version 24 or newer.
- **CON-DELIV-01 (Delivery):** The system shall be delivered as a single executable .jar file requiring no installation steps beyond a compatible JRE.
- **CON-INPUT-01 (Input):** The game shall be fully operable using a mouse only during gameplay; keyboard use is permitted only for text entry in name, K value, and IP address fields.
- **CON-NET-01 (Network):** LAN play shall operate over a local area network only; no internet connectivity is required or assumed.

4 Functional Requirements

4.1 Use Cases

UC-01 Configure and Start Local Game

- **ID:** UC-01
- **Description:** A player opens the application, selects local game mode, enters K, selects a deck size from the computed options, optionally enters player names, and starts a new local game session.
- **Actor:** Player
- **Preconditions:** Application is launched; main menu is displayed.
- **Postconditions:** A local game session is initialised with the chosen configuration; the board is displayed and both players are identified by name.
- **Successful Course:** Player selects local game, enters a value for K (1–45), the system computes and displays up to three valid deck size options as multiples of K not exceeding 45, the player selects one, optionally enters names (defaults to Player 1 and Player 2), and the game board is shown.
- **Abnormal Course:** Player enters a K value outside the valid range; system displays a validation error and prompts re-entry.
- **Priority:** High
- **Referenced Requirements:** FR-MENU-01, FR-MENU-03, FR-MENU-04, FR-MENU-05, FR-VAL-01, FR-VAL-02

UC-02 Host a LAN Game

- **ID:** UC-02
- **Description:** A player chooses to host a LAN game, enters their name, configures K and deck size, and waits for the other player to connect.
- **Actor:** Host player
- **Preconditions:** Application is launched; main menu is displayed.
- **Postconditions:** A LAN session is open and the host is waiting for a client connection.
- **Successful Course:** Host selects “Host LAN Game”, enters their player name, configures K and selects a deck size, and the system displays the host machine’s IP address with a message indicating it is waiting for the other player to join.
- **Abnormal Course:** The system cannot determine a local IP address; an error message is displayed and the host is returned to the main menu.
- **Priority:** High
- **Referenced Requirements:** FR-LAN-01, FR-LAN-02, FR-LAN-03

UC-03 Join a LAN Game

- **ID:** UC-03
- **Description:** A player chooses to join an existing LAN game by entering the host’s IP address and their own player name.
- **Actor:** Client player
- **Preconditions:** A host machine is already waiting for a connection; the client knows the host’s IP address.
- **Postconditions:** Connection is established; both machines transition to the lobby screen.
- **Successful Course:** Client selects “Join LAN Game”, enters the host’s IP address and their own player name, and clicks Connect. The system establishes the connection and both machines display the lobby screen.
- **Abnormal Course:** The entered IP address is unreachable or the connection is refused; the system displays an error message and allows the client to re-enter the

IP address or return to the main menu.

- **Priority:** High
- **Referenced Requirements:** FR-LAN-04, FR-LAN-05, FR-LAN-06

UC-04 LAN Lobby and Game Start

- **ID:** UC-04
- **Description:** After a successful LAN connection, both players see a lobby screen confirming the connection before the game begins automatically.
- **Actor:** Both players
- **Preconditions:** LAN connection between host and client is established.
- **Postconditions:** Both players are shown the game board and the game session begins.
- **Successful Course:** Both machines display a lobby screen showing both player names and a connected status message. After a brief fixed display period, the game board is shown automatically on both machines and the game begins.
- **Abnormal Course:** Connection drops during the lobby screen; both players are returned to the main menu with an error message.
- **Priority:** High
- **Referenced Requirements:** FR-LAN-06, FR-LAN-07, FR-LAN-08

UC-05 Flip Cards and Match

- **ID:** UC-05
- **Description:** The active player clicks cards one at a time to flip them and attempt to form a K-tuple match.
- **Actor:** Active player
- **Preconditions:** A game session is active; it is the actor's turn; at least K face-down cards remain.
- **Postconditions:** Either a match is recorded and the active player continues their turn, or a mismatch occurs and the turn passes to the opponent.
- **Successful Course:** Player clicks K face-down cards with matching symbols; cards are removed from the board; player's score increases by 10 points; the turn remains with the same player and they may continue flipping.
- **Abnormal Course:** K flipped cards do not all match; cards remain face-up for `MISMATCH_DELAY_MS` then flip back face-down; turn passes to the opponent.
- **Priority:** High
- **Referenced Requirements:** FR-FLIP-01, FR-FLIP-02, FR-FLIP-03, FR-FLIP-04, FR-TURN-01, FR-TURN-02, FR-TURN-03, FR-SCORE-01, FR-SCORE-02

UC-06 End Game and View Result

- **ID:** UC-06
- **Description:** When all cards have been matched the game ends and both players see the final result screen.
- **Actor:** Both players
- **Preconditions:** All cards have been removed from the board.
- **Postconditions:** Final scores and outcome are displayed; players may start a new game or return to the main menu.
- **Successful Course:** System detects no remaining cards, displays final scores and outcome (win / lose / draw), and offers options to start a new game or return to the main menu.

- **Abnormal Course:** Score data fails to synchronise between LAN clients; system displays locally computed result and flags the discrepancy.
- **Priority:** High
- **Referenced Requirements:** FR-SCORE-03, FR-SCORE-04, FR-SCORE-05

UC-07 Handle Mid-Game Disconnection

- **ID:** UC-07
- **Description:** The LAN connection between the two clients is lost during an active game session.
- **Actor:** Both players (system-triggered)
- **Preconditions:** A LAN game session is active; the network connection is interrupted.
- **Postconditions:** Both clients display a disconnection error and the game session is terminated.
- **Successful Course:** The system detects the lost connection, immediately halts gameplay on both clients, displays a disconnection message, and offers both players the option to return to the main menu.
- **Abnormal Course:** Only one client detects the disconnection; that client displays the error and returns to the main menu independently.
- **Priority:** High
- **Referenced Requirements:** FR-NET-01, FR-NET-02

4.2 Detailed Functional Requirements

ID	Description	Prio
<i>Main Menu and Configuration</i>		
FR-MENU-01	The system shall display a main menu on startup with the following options: start a local game, open the LAN game section, and exit the application.	High
FR-MENU-02	The LAN game section of the main menu shall offer two options: Host LAN Game and Join LAN Game.	High
FR-MENU-03	The system shall allow the player to enter a name for each player before starting a game. If no name is entered, the system shall default to “Player 1” and “Player 2”.	High
FR-MENU-04	The system shall prompt the player to enter a value for K before any deck size is presented. K shall be a positive integer in the range 1–45. If the entered value is outside this range or is not a positive integer, the system shall display a validation error and require re-entry before proceeding.	High
FR-MENU-05	After a valid K has been entered, the system shall compute all positive multiples of K that do not exceed 45 and present up to three of these values as selectable deck size options. If more than three valid multiples exist, the system shall present the three largest. The player must select one of the offered options before the game can be started.	High

FR-VAL-01	The system shall enforce the constraint <i>deck size mod K = 0</i> so that every card in the deck belongs to exactly one valid K-tuple. A game shall not be startable with a combination that violates this constraint.	High
FR-VAL-02	The deck size options presented to the player shall always satisfy <i>deck size mod K = 0</i> by construction (see FR-MENU-05), so no additional validation error for this constraint is required at confirmation time.	High
FR-VAL-03	The number of distinct symbols in the deck shall equal <i>deck size / K</i> . Where this value is less than 15 due to the chosen K and deck size, the system shall still permit the game to start; the 15-symbol count is a target lower bound, not an absolute restriction (see Section 1.3).	Medium
<i>LAN Connection Flow</i>		
FR-LAN-01	When the host selects “Host LAN Game”, the system shall prompt the host to enter their player name and configure K and deck size before opening the waiting screen.	High
FR-LAN-02	After the host completes configuration, the system shall display the host machine’s local IP address and a status message indicating that it is waiting for the other player to join.	High
FR-LAN-03	The host shall be able to cancel the waiting state and return to the main menu at any point before a client connects.	Medium
FR-LAN-04	When the client selects “Join LAN Game”, the system shall present a form allowing the client to enter the host’s IP address and their own player name.	High
FR-LAN-05	The system shall attempt to establish a connection to the entered IP address when the client clicks Connect. If the connection cannot be established, the system shall display a descriptive error message and allow the client to correct the IP address or return to the main menu.	High
FR-LAN-06	Upon successful connection, both machines shall transition to the lobby screen.	High
FR-LAN-07	The lobby screen shall display both player names and a message confirming that the connection has been established.	High
FR-LAN-08	The lobby screen shall be displayed for a fixed period (default 3 seconds) after which both machines shall automatically transition to the game board and the game session shall begin.	High
<i>Gameplay</i>		
FR-FLIP-01	The system shall flip a face-down card to reveal its symbol image when the active player clicks on it.	High
FR-FLIP-02	When K face-up cards share the same symbol (a match), the system shall remove those cards from the board, award 10 points to the active player, and keep the turn with the same player so they may continue flipping.	High

FR-FLIP-03	When K face-up cards do not all share the same symbol (a mismatch), the system shall keep the cards face-up for a duration defined by the constant <code>MISMATCH_DELAY_MS</code> (default 1500 ms), then flip all K cards back face-down and pass the turn to the opponent.	High
FR-FLIP-04	Only face-down cards shall be selectable. Cards that have already been removed (matched) or are currently face-up within an active K-tuple selection shall not respond to click events.	High
FR-BOARD-01	Cards shall be arranged in a rectangular grid layout on the game board. The layout shall scale appropriately to fit the selected deck size on screen.	High
FR-TURN-01	The system shall randomly determine which player takes the first turn at the start of each game session. The selected starting player shall be shown to both players before the first move.	High
FR-TURN-02	The system shall display the active player's name clearly on screen at all times during gameplay so both players know whose turn it is.	High
FR-TURN-03	During LAN play, every card flip event performed by the active player shall be broadcast to the remote client so that both players see an identical board state at all times.	High
FR-TURN-04	During LAN play, the inactive player's client shall not accept card-click input; the board shall be read-only until the turn changes.	High
<i>Scoring and End of Game</i>		
FR-SCORE-01	Each successful K-tuple match shall award 10 points to the active player. The player with the most points when no cards remain wins the game.	High
FR-SCORE-02	The current score of both players shall be visible on screen throughout gameplay.	High
FR-SCORE-03	When all cards have been removed from the board, the system shall end the game and display a result screen showing each player's name, their final score, and the outcome: win, lose, or draw. A draw occurs when both players have equal final scores.	High
FR-SCORE-04	The result screen shall offer the player two options: start a new game with the same configuration, or return to the main menu.	High
FR-SCORE-05	When the player selects "New Game" from the result screen, the system shall reset the board and scores and start a fresh game session with the same players, K value, and deck size without returning to the configuration screens.	High
<i>Network Error Handling</i>		
FR-NET-01	The system shall detect a loss of LAN connection during an active game session within 5 seconds of the disconnection occurring.	High

FR-NET-02	Upon detecting a disconnection, the system shall immediately halt gameplay on the affected client, display a disconnection error message, and offer the player the option to return to the main menu.	High
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5 Quality Requirements

5.1 Usability

- **QR-USAB-01:** The system shall display the active player's name and both players' current scores at all times during gameplay. **Reference:** FR-TURN-02, FR-SCORE-02.
Fit criterion: In 10 observed game sessions, both players can identify the active player and both scores without additional explanation.
- **QR-USAB-02:** The deck size option list shall update immediately and correctly whenever the player changes the value of K. **Reference:** FR-MENU-04, FR-MENU-05.
Fit criterion: In 20 observed configuration attempts with varying K values, the displayed deck size options are always valid multiples of the entered K not exceeding 45.
- **QR-USAB-03:** A first-time user shall be able to navigate from the main menu to a running game — both local and LAN — without external assistance.
Fit criterion: In 5 observed first-time attempts per game mode, all users successfully start a game without help.
- **QR-USAB-04:** The LAN host waiting screen shall clearly display the IP address the joining player must enter. **Reference:** FR-LAN-02.
Fit criterion: In 10 observed LAN setup attempts, all joining players correctly identify and enter the IP address without assistance.
- **QR-USAB-05:** The game board shall have a regular, clean visual appearance appropriate for an office environment.
Fit criterion: In an informal review by 3 stakeholders, all rate the visual appearance as acceptable for workplace use.

5.2 Performance

- **QR-PERF-01:** A card flip shall be visible within 500 ms of the active player's click during local play. **Reference:** FR-FLIP-01.
Fit criterion: In 30 flip events, the flip is visible within ≤ 500 ms each time.
- **QR-PERF-02:** During LAN play, a card flip shall be visible on the remote client within 500 ms of the active player's click. **Reference:** FR-TURN-03.
Fit criterion: In 30 flip events measured over a local network, the flip is visible on the remote client within ≤ 500 ms each time.

- **QR-PERF-03:** Match detection and card removal shall complete within 1 second after the K-th matching card is flipped. **Reference:** FR-FLIP-02.
Fit criterion: In 20 successful matches, cards are removed and score is updated within ≤ 1.0 s each time.
- **QR-PERF-04:** After a mismatch, all K mismatched cards shall flip back face-down within 500 ms of the `MISMATCH_DELAY_MS` period expiring. **Reference:** FR-FLIP-03.
Fit criterion: In 20 observed mismatches, all cards are face-down and the turn has passed within $\leq (\text{MISMATCH_DELAY_MS} + 500)$ ms of the mismatch being detected each time.
- **QR-PERF-05:** The final result screen shall appear within 2 seconds of the last card being removed. **Reference:** FR-SCORE-03.
Fit criterion: In 10 completed games, the result screen is displayed within ≤ 2.0 s each time.

5.3 Reliability

- **QR-RELY-01:** The game shall complete a full session from start to result screen without crash or data loss.
Fit criterion: 20 consecutive complete game sessions finish without application crash and with a correct final result on screen.
- **QR-RELY-02:** During LAN play, board state shall remain identical on both clients throughout a game session. **Reference:** FR-TURN-03.
Fit criterion: In an audit of 10 complete LAN sessions, the sequence of card states recorded on both clients is identical in 10/10 sessions.
- **QR-RELY-03:** Score increments shall correspond exactly to successful matches. **Reference:** FR-FLIP-02, FR-SCORE-02.
Fit criterion: In an audit of 50 match events, each score increment of 10 points corresponds to exactly one valid K-tuple removal.
- **QR-RELY-04:** A mid-game LAN disconnection shall be detected and communicated to the affected player within 5 seconds. **Reference:** FR-NET-01, FR-NET-02.
Fit criterion: In 10 simulated disconnection events during active LAN sessions, the error message is displayed on the affected client within ≤ 5 s each time.

5.4 Maintainability

- **QR-MAIN-01:** The mismatch delay duration shall be defined as the named constant `MISMATCH_DELAY_MS` in the source code so that it can be adjusted without modifying game logic. **Reference:** FR-FLIP-03.
Fit criterion: Changing `MISMATCH_DELAY_MS` and rebuilding produces the expected delay in all mismatch scenarios without any other code changes.

- **QR-MAIN-02:** The project shall build with a standard Maven or Gradle command and provide reproducible run instructions in a README file.
Fit criterion: Build succeeds from a clean checkout and the README contains a working build and run section.
- **QR-MAIN-03:** Automated tests shall cover K input validation, deck size option computation, match detection, mismatch and flip-back logic, and score calculation.
Fit criterion: At least one automated test references each of FR-MENU-04, FR-MENU-05, FR-FLIP-02, FR-FLIP-03, FR-SCORE-01.

6 Acceptance Criteria

The delivered system is considered acceptable when all of the following conditions are met:

- All High-priority functional requirements are implemented and verifiable.
- All constraints (CON-TECH-01, CON-DELIV-01, CON-INPUT-01, CON-NET-01) are satisfied.
- No listed Quality Requirement fit criterion is violated.
- K can be set to any integer in the range 1–45, and the system presents only valid multiples of K as deck size options (FR-MENU-04, FR-MENU-05).
- A match keeps the turn with the active player; a mismatch passes the turn to the opponent (FR-FLIP-02, FR-FLIP-03).
- Player names are displayed correctly on the lobby screen, throughout gameplay, and on the result screen (FR-LAN-07, FR-MENU-03, FR-SCORE-03).
- The result screen offers both a new game option and a return-to-menu option (FR-SCORE-04, FR-SCORE-05).
- The LAN host waiting screen displays the correct local IP address and the client can successfully connect by entering it (FR-LAN-02, FR-LAN-05).
- Both clients display an identical board state throughout a complete LAN game session (FR-TURN-03, QR-RELY-02).
- A mid-game LAN disconnection is detected and the error is shown to the player within 5 seconds (FR-NET-01, FR-NET-02).
- A draw outcome is correctly detected and displayed when both players finish with equal scores (FR-SCORE-03).